

PRODUCT REQUIREMENTS DOCUMENT

Village Retail Store — Smart Pre-Order Web App

Version 4.0 | May 2026

Field	Details
Version	4.0 (revised from 3.0)
Date	May 2026
Owner	Store Owner
Primary Language	Hinglish (Single language — see Section 10)
Target Region	Rajasthan, India
Platform	Mobile-First Progressive Web App (PWA)
Target Users	Max 100 customers, 1 owner
Budget Constraint	Zero recurring cost — Free tier only

v4.0 Changes: This version incorporates architectural fixes, edge case handling, simplified systems for 100-user scope, WhatsApp-based free OTP, and 6 new features. Changes are marked [UPDATED] or [NEW] throughout.

1. Executive Summary

A lightweight, mobile-first Progressive Web App designed for Indian village retail stores. Customers can browse products, place pre-orders, track orders, maintain khata (credit), and reorder frequently purchased groceries before visiting the store.

The app solves real operational problems for a local dukaan owner: chaotic walk-in traffic, repeated verbal orders, manual khata tracking, uncertain customer arrivals, and lack of order history.

Optimized for:

- Simplicity and low-data usage
- Repeat grocery ordering with Hinglish UI
- Fast pickup workflow for village customers
- Zero recurring cost — runs entirely on Firebase free tier
- Fully manageable by a single owner using only a smartphone

2. Product Vision

"A Smart Digital Dukaan Assistant for Village Retail"

This product is NOT intended to become a grocery marketplace, a delivery platform, or a multi-store ecommerce application.

Focused on:

- Faster order preparation and repeat customer convenience
- Simplified digital khata management
- Customer trust and loyalty
- Reduced manual effort for the owner
- Digital ordering habit formation in village customers

3. Problem Statement

Current Situation

Customers currently visit the store without knowing if products are available, whether the order can be packed quickly, or how long pickup may take.

The owner manually listens to verbal orders, packs items while customers wait, maintains khata records in notebooks, and tracks regular customer preferences mentally.

This creates customer wait times, operational confusion, missed sales, repeated manual work, and no digital record of demand patterns.

4. Goals & Success Metrics

Goal	Success Metric
------	----------------

Increase pre-orders	30%+ regular customers place at least one order monthly
Reduce pickup waiting time	Average pickup under 5 minutes
Reduce owner workload	Lower verbal/manual order dependency
Improve customer retention	Customers actively use Buy Again and Usual Orders
Reduce fake/no-show orders	Customer label system active
Simplify khata management	Replace physical khata notebook
Increase repeat ordering	Frequent customers reorder digitally
Zero recurring cost	App runs entirely on Firebase Spark (free) plan

5. Target Users

Persona 1 — Village Customer

Attribute	Details
Age	20–55
Device	Android smartphone
Internet	4G / unstable mobile internet
Language	Hinglish (primary)
Tech Familiarity	Uses WhatsApp daily
Goal	Place order quickly and pick up without waiting
Max Users	100 registered customers

Persona 2 — Store Owner

Attribute	Details
Age	30–50
Device	Android smartphone
Technical Skill	Moderate — uses WhatsApp, basic apps
Goal	Pack orders before customer arrives
Pain Point	Manual work and repeated interruptions
Count	1 owner, no staff accounts in MVP

6. Product Principles

The application must prioritize:

- Simplicity over feature complexity
- Fast ordering over visual richness
- Familiar Indian mobile UX patterns (WhatsApp-like flows)
- Zero recurring cost — Firebase Spark free plan only, no Cloud Functions in MVP
- Hinglish as the single language — no language toggle in MVP

- Repeat customer convenience with minimal typing
- Smartphone-first usability

Removed: Language toggle deferred to Phase 2. Single Hinglish mode reduces translation maintenance by 50% and village users will not meaningfully use a Hindi/Hinglish toggle.

7. Technical Architecture

UPDATED: Cloud Functions removed from MVP to maintain zero cost on Firebase Spark plan. All business logic moves to client-side with strict Firestore Security Rules.

[Customer PWA] ↔ [Firestore + Firebase Auth] ↔ [Owner Dashboard PWA]

Business logic resides in:

- Firestore Security Rules — access control and validation
- Client-side React logic — cart totals, order state, trust label display
- Firebase Hosting — static PWA hosting (free tier)
- Firebase Auth — phone number authentication

Cloud Functions will be added in Phase 2 only if a specific need arises that cannot be handled client-side.

8. Technology Stack

Layer	Technology	Cost
Frontend	React PWA	Free
Backend Logic	Firestore Security Rules + Client JS	Free (no Cloud Functions)
Database	Firestore (Spark plan)	Free up to 1GB / 50k reads / 20k writes per day
Authentication	Firebase Auth — Phone Number (WhatsApp OTP)	Free
Notifications	Firebase Cloud Messaging (FCM) + WhatsApp fallback	Free
Hosting	Firebase Hosting (Spark)	Free up to 10GB/month
Fonts	Noto Sans Devanagari	Free (Google Fonts)
OTP Channel	WhatsApp Business API (free tier) via wati.io or similar	Free tier

Cost Note: Firebase Spark plan is entirely free and handles 100 users comfortably. Estimated Firestore usage at 100 users / 20 orders/day is well within free limits. No paid plan needed for MVP.

9. Authentication & Security

[UPDATED] Authentication

UPDATED: Switched to mobile number as primary identifier (matching Indian app norms — GPay, Meesho, PhonePe). Username/password removed. WhatsApp OTP used instead of SMS to eliminate all OTP costs.

Registration Flow

Required fields:

- Full Name
- Mobile Number (primary identifier)
- WhatsApp OTP verification (only during initial signup)

No username or password required. Mobile number is the login identity. This matches what village users already do on WhatsApp, Paytm, and GPay.

Login Flow

- Enter mobile number → receive WhatsApp OTP → logged in
- Session persists for 30 days (no repeated OTP on same device)
- OTP only required on new device or session expiry

OTP Channel — WhatsApp (Free)

OTP is sent via WhatsApp message, not SMS. WhatsApp Business API free tier supports this at zero cost. Recommended provider: wati.io or Twilio WhatsApp Sandbox (both have free tiers for low volume).

Why WhatsApp OTP: Village users already have WhatsApp on their phone. A WhatsApp message is more visible than an SMS, costs nothing, and does not require a paid SMS gateway.

Security Requirements

- Session persistence — stay logged in for 30 days on same device
- Rate limiting via Firebase Auth built-in protections (no custom rate limiting needed for 100 users)
- Firestore Security Rules — users can only read/write their own data
- Owner account — separate role flag in Firestore user document

Rule	Limit
OTP requests	Max 5/hour (Firebase Auth built-in)
Session duration	30 days before re-authentication
Owner account	Single owner UID — hardcoded in security rules

10. Language Strategy

[UPDATED] Language

UPDATED: Language toggle removed for MVP. Single Hinglish mode only. This halves translation work and village users will not meaningfully use a Hindi/Hinglish toggle. Toggle can be added in Phase 2 based on actual feedback.

Application Language: Hinglish

All UI copy is in Hinglish — a natural mix of Hindi and English that village customers use daily. Examples: "Order karo", "Abhi available hai", "Mera khata", "Dobara order karo".

Product Naming Model

Each product stores multiple name variants to support search across languages:

- Hinglish display name: "Sarson ka Tel"
- Hindi name: "सरसों का तेल"
- English aliases: "Mustard Oil", "mustard"
- Search keywords: "tel", "sarson", "mustard"

Search Requirements

Search must support Hindi input, Hinglish input, English input, and spelling variations. Example: daal / dal / दाल — all should return the correct product.

Order Notes

Order notes accept any input — Hinglish, Hindi Devanagari, or English. The keyboard is whatever the customer's Android default is. Example note: "Kam mirch wala dalna" or "कम मिर्च वाला दालना".

Removed: The PRD v3 example note was in pure Devanagari which requires keyboard switching. Hinglish roman input is the natural behavior for this audience.

11. Product Catalog

Product Display Fields

Field	Notes
Product Name	Hinglish display name
Variants	Size/weight options (e.g., 1kg, 2kg, 5kg) — NEW, see below
Price per variant	Set per variant, not per product
Unit	kg, litre, packet, piece
Category	From fixed category list
Availability Status	Available Today / Running Low / Out of Stock

[UPDATED] Product Variants — Critical Addition

NEW: Product variants are required for grocery catalogs. Without variants, the owner must create separate catalog entries for Atta 1kg, Atta 2kg, Atta 5kg — bloating the catalog and breaking search. Variants solve this at the data model level.

Variant Data Structure

Each product has an optional variants array. If variants exist, the customer selects a variant after tapping the product. If no variants, quantity is entered directly.

Example product structure:

```
{ "name": "Atta", "variants": [ { "label": "1 kg", "price": 45 }, { "label": "2 kg", "price": 88 }, { "label": "5 kg", "price": 210 } ] }
```

Availability States

The application does NOT show exact inventory quantities. Instead:

- Available Today — normal ordering enabled
- Running Low — order enabled with caution message: "Jaldi karo, sirf thoda bacha hai"
- Out of Stock — ordering disabled for this product

This matches real village grocery behavior and reduces inventory maintenance complexity.

12. Product Categories

Category	Example Products
Atta & Grains	Gehun Atta (1kg, 2kg, 5kg, 10kg), Chawal, Maida, Besan
Oil	Sarson Tel, Refined Tel, Soyabean Tel, Tili Tel (500ml, 1L, 5L)
Masale	Lal Mirch, Haldi, Dhaniya, Garam Masala, Namak
Snacks	Namkeen, Biscuits, Chips
Soap & Shampoo	Lifebuoy, Lux, Head & Shoulders
Daily Use	Agarbatti, Matchbox, Diya
Dairy	Doodh, Dahi, Paneer
Cold Drinks	Pepsi, Frooti, Limca (250ml, 600ml)

13. Catalog UX Features

Search Bar

- Accepts Hinglish, Hindi Devanagari, and English input
- Searches product name, variants, keywords, and aliases
- Auto-shows suggestions after 2 characters

Category Tabs

- Horizontal scrollable category navigation at top of catalog
- Tapping a category filters the product list

Recently Ordered

- Shows items from the customer's last 3 orders
- One-tap add to cart

Frequently Bought

- Shows the customer's 5 most reordered products

Your Usual Order (Aapka Usual)

- System-generated suggestion based on customer order history
- Example: "2kg Atta, 1L Sarson Tel, 1kg Cheeni" — Ek click mein add karo

14. Cart System

Cart Features

- Add / remove quantity per item
- Variant selection per item (e.g., 2kg vs 5kg Atta)
- Cart persistence — saved to Firestore, survives app close
- Line item totals and grand total
- Order notes field (accepts Hinglish / Hindi / English text)

[UPDATED] Minimum Order Value

NEW: A minimum order value prevents trivial orders that waste owner packing time.

- Default minimum order: Rs. 50
- Owner can configure this from the dashboard (range: Rs. 20 to Rs. 500)
- Cart shows "Minimum order Rs. 50" when total is below threshold
- Order placement is disabled until minimum is met

Price Snapshot at Order Placement

- Item price is captured (snapshotted) at the moment of order placement
- If owner updates prices in the morning before opening, all subsequent orders use the new price

- Cart shows a "Price updated, please review" banner if price changes while item is in cart

Note: Owner updates prices once in the morning — this is the agreed workflow. No complex price-change conflict resolution needed.

15. Pickup Slots

[UPDATED] Pickup Slots

UPDATED: The '5–10 minute' slot is only shown when the store is open. When store is closed, it is hidden entirely. This prevents the slot from creating false promises.

Slot	Visibility Condition
5–10 minutes	Only shown when store is marked Open by owner
30 minutes	Always visible when store is open
1 hour	Always visible when store is open
By 5 PM	Always visible when store is open

No dynamic slot capacity logic required in MVP.

[NEW] 16. Store Open / Closed Toggle

NEW: This is a critical missing feature from v3.0. Without it, customers can place orders at midnight or when the owner is unavailable, creating broken experiences.

Owner Control

- Owner dashboard has a prominent toggle: Store Khuli Hai / Dukaan Band Hai
- Toggling closed immediately hides the order placement button for all customers
- Owner can set a reopen message: "Kal subah 9 baje kholenge"

Customer Experience

- When store is closed, catalog is still browsable (offline-cache works)
- Order button is replaced with: "Dukaan abhi band hai — [owner message]"
- 5–10 minute pickup slot is hidden when store is closed

Auto-Close (Optional)

- Owner can set operating hours (e.g., 8 AM – 8 PM)
- App automatically shows closed state outside those hours
- Owner can still manually override to open/close at any time

17. Order Management System

Order Statuses

Placed	□ □ □ □ □ □ □ □ □ □
Being Packed	□ □ □ □ □ □ □ □ □ □
Ready for Pickup	□ □ □ □ □ □ □ □ □ □
Completed	□ □ □ □ ✓
Cancelled	□ □ □ □

Customer Order Features

- Place orders, view order history, track current order status
- Cancel pending orders (before packing starts)
- Add items to a pending order — allowed before owner starts packing
- Reorder previous orders (Buy Again)
- View khata balance

NEW: Customers can add items to an existing Placed order before packing begins. Once owner moves order to 'Being Packed', the order is locked. This prevents the punishing cancel-and-reorder cycle.

Duplicate Order Prevention

- Customer cannot place a second new order while one order is in Placed or Being Packed status
- If customer wants different items, they use 'Add to Order' on the existing order
- Once order reaches Ready for Pickup, customer can place a new order

18. Partial Order Handling

[UPDATED] Partial Order Handling

UPDATED: v3.0 did not define behavior when ALL items in an order become unavailable. This is now specified.

Some Items Unavailable

- Owner removes unavailable items from the order
- Remaining order continues normally
- Total updates automatically
- Customer receives push notification: "Aapke order mein se [item] nahi mila — baki order tayyar ho raha hai"

ALL Items Unavailable (New)

- If owner removes all items, the order is auto-cancelled
- Customer receives notification: "Aaj aapka order available nahi hai — sorry!"
- Khata is NOT affected — no amount is added for a fully cancelled order
- Trust score is NOT reduced — this is a store-side cancellation, not customer fault

19. Buy Again & Repeat Ordering

Buy Again

- One tap re-adds all items from a previous order into cart
- Customer can review and modify before placing

Frequently Bought

- System automatically surfaces the customer's most ordered products
- Displayed prominently on catalog home screen

Usual Orders

- App suggests a regular grocery combination based on customer order history
- Shown as: "Aapka Usual: 2kg Atta, 1L Tel, 1kg Cheeni — Sab add karo?"
- Customer can accept the entire suggestion or modify before placing

20. Customer Trust Label System

[UPDATED] Trust System — Simplified for 100 Users

UPDATED: The algorithmic trust score (0–100 points, complex tier rules) has been replaced with a simple owner-assigned label system. For 100 known village customers, the owner knows each person. A computed score adds complexity without proportional value.

Owner-Assigned Customer Labels

Label	Color	Meaning
Trusted	Green	Reliable customer, always picks up
Normal	Gray	Default label for all new customers
Careful	Amber	History of cancellations or no-shows

Default for New Customers

- All new customers start with the Normal label
- Owner can change a customer's label at any time from the customer profile in dashboard
- No automatic label changes — owner decides based on their real-world knowledge

Owner Dashboard Visibility

- Each order card shows the customer's label badge
- If a Careful customer places an order, owner sees an alert: "Dhyan dena — pehle no-show tha"

Order History per Customer

- Owner can view any customer's profile showing: total orders, cancellations, no-shows, khata balance
- This gives the owner the context to assign labels accurately

Future (Phase 2): Automatic scoring based on behavior patterns can be added once the app has 3+ months of real data.

21. Abuse Prevention System

Login Protection

- Firebase Auth built-in rate limiting and reCAPTCHA
- No custom rate limiting code needed for 100 users

Order Protection

- Maximum 1 active order at a time per customer
- Careful-labeled customers trigger owner alert on new order
- Cancel before packing allowed freely — no punishment for legitimate order changes

Owner Visibility

- Customer profile shows full order history including cancellations and no-shows
- Owner can see: Label, Total Orders, Cancellations, Khata Balance on each order card

22. Khata (Credit) System

[UPDATED] Khata — Two Additions

UPDATED: Two new features added: (1) Per-customer credit limit to prevent unchecked debt accumulation. (2) Customer dispute flag so customers can report incorrect entries.

Khata Features

- Running balance per customer
- Full ledger history with date, type, amount, note
- Partial payments and payment notes
- Settlement tracking
- Manual adjustment support by owner

Credit Limit (New)

- Each customer has a configurable credit limit (default: Rs. 500)
- Owner can change the limit per customer from the dashboard
- When a customer's outstanding khata balance reaches their limit, new orders using khata are blocked
- Customer sees: "Aapka khata Rs. 500 ho gaya — pehle thoda pay karo"
- Owner sees an alert on the order card: "Khata limit pe hai"

Dispute Flag (New)

- Customer can tap "Galat lag raha hai" on any khata entry
- This sends the owner a notification: "[Customer name] ne entry flag ki hai — check karo"
- No complex dispute resolution flow — owner resolves manually and can update/delete entries
- Builds customer trust by giving them a voice on their own account

Example Ledger

Date	Type	Amount	Note
2 May	Order #12	+Rs. 240	Grocery order
5 May	Payment	-Rs. 100	Cash paid
7 May	Order #15	+Rs. 180	Usual order
Balance		Rs. 320	Due

Customer View

- Current due balance (prominently displayed)
- Full ledger history
- Settled amounts
- Flag entry button on each row

23. Owner Dashboard

Home Screen

- Store Open / Closed toggle — most prominent element
- Today's order count and pending orders
- Simple summary: Today's total sales, total due khata

Product Management

- Add, edit, delete products and variants
- Update prices per variant
- Change categories
- Mark as Available / Running Low / Out of Stock

Order Dashboard

Each order card displays:

- Customer name and phone number
- Customer label badge (Trusted / Normal / Careful)
- Items list with variants and quantities
- Order total
- Pickup slot selection
- Order status with action buttons: Pack → Ready → Done
- Owner note field — to send a message to the customer about their order

Customer Management

- List of all registered customers
- Each customer profile: label, order history, cancellations, khata balance
- Owner can change customer label, update credit limit, add manual khata entries

24. Notifications System

[UPDATED] Notifications — WhatsApp Fallback Added

UPDATED: FCM push notifications on Android PWAs can be suppressed by battery saver mode. WhatsApp notification added as a reliable fallback for the owner, using the free WhatsApp Business API.

Technology

- Primary: Firebase Cloud Messaging (FCM) — push notifications via PWA
- Owner fallback: WhatsApp Business API — if FCM fails or owner has not opened PWA

Customer Notifications (FCM)

Trigger	Message (Hinglish)
Order placed	Aapka order mil gaya! Tayar hote hi batayenge.
Being packed	Aapka order pack ho raha hai!
Ready for pickup	Aapka order ready hai — aa jao!
Order modified (items removed)	Kuch items available nahi the — baki order ready hai.
Order cancelled by store	Aaj aapka order available nahi — sorry!
Khata entry flagged response	Owner ne aapki entry check kar li hai.

Owner Notifications

Trigger	FCM	WhatsApp Fallback
New order placed	Yes	Yes — if not acknowledged in 5 min
Order cancellation	Yes	No
Careful customer order	Yes — with alert badge	No
Khata entry flagged by customer	Yes	No
Khata credit limit reached	Yes	No

[NEW] 25. Store Discovery & Customer Onboarding

NEW: v3.0 had no mechanism for how customers find or install the PWA. This is a critical gap — without discoverability, adoption never happens.

Store QR Code

- Owner dashboard generates a unique store QR code
- Owner prints and sticks the QR code at the store counter
- Scanning the QR code opens the PWA directly in the customer's browser
- First-time visitors see the registration screen; returning users are auto-logged in

WhatsApp Share Link

- Owner can copy a shareable link from the dashboard
- Owner sends this link to customers on WhatsApp: "Hamare dukaan ki app download karo, order easy ho jayega!"
- Link opens the PWA on mobile — no App Store needed

PWA Install Prompt

- First visit shows: "Hone Dukaan apne phone mein add karo" with Add to Home Screen prompt
- This is a standard PWA feature — no app store submission needed

[NEW] 26. Store Info Page

[NEW]: Customers need a way to find store address, phone, and hours from within the app.

- Store name and owner name
- Address (text field)
- Phone number (tappable — opens dialer)
- Operating hours
- "Raste mein" button — opens Google Maps with store address

Owner configures store info during initial setup. Editable from dashboard settings.

[NEW] 27. Owner-to-Customer Order Notes

NEW: Owner needs a way to communicate substitutions or questions without calling the customer. Example: 'Sarson tel nahi hai — refined chalega?'

- Owner can add a note to any order in the dashboard
- Customer receives a push notification and sees the note on their order status screen
- Customer can respond with one of two taps: Theek Hai (OK) or Nahi Chahiye (Cancel this item)
- Simple binary response — no free-text chat to keep it simple

28. Analytics (Simplified)

[UPDATED] Analytics — Simplified for 100 Users

UPDATED: Full analytics dashboard removed for MVP. For 10–20 orders/day from 100 known customers, statistical analytics are not actionable. Replaced with simple summary cards that are genuinely useful.

Owner Home Screen Summary Cards

- Today's orders — count and total value
- Pending orders — needs action
- Total khata outstanding — across all customers
- This week's sales — simple total

Phase 2 Analytics (Not in MVP)

- Top 5 products this month
- Peak order day/time
- Cancelled orders tracking
- Customer lifetime value

29. Reward Coins System

[UPDATED] Reward Coins — Removed from MVP Scope

UPDATED: Coins architecture removed from MVP entirely. Firestore is schema-less — adding a coins field takes 10 minutes when Phase 3 begins. Pre-building unused schema adds dead code and cognitive overhead.

Planned for Phase 3:

- Rs. 100 order = 1 coin
- Festival offers, loyalty rewards, discounts
- Customer retention campaigns

Not in MVP schema. Will be added as a separate Firestore collection in Phase 3.

30. Performance Requirements

[UPDATED] Performance Requirements — Scope Corrected

Requirement	Target	Notes
Initial Load	Under 3 seconds on 4G	
Low Data Usage	Under 500KB/session	
Device Support	Android 8+	Primary target audience
Offline Catalog	Cached browsing via Service Worker	Orders/khata require active internet
Registered Users	Up to 500 registered accounts	
Concurrent Active Users	Up to 100 at any time	Corrected from v3.0 which said 500
Daily Backup	Firestore automatic backup	
Offline Behavior	Catalog browsable offline. Cart, orders, and khata require internet. No background sync.	

Corrected: v3.0 listed '500 users' as scalability target but the business context is 100 users. This caused a contradiction. 500 registered accounts is a realistic upper bound for a growing village dukaan.

31. Non-Goals (Not Included in MVP)

- Home delivery
- Online payment gateway (UPI/Razorpay — Phase 3)
- Dynamic real-time inventory quantities
- Queue visibility for customers
- Multi-store support
- Staff / employee roles
- Export systems (CSV, reports)
- Advanced offline sync / background sync
- Language toggle (Phase 2)
- Reward coins (Phase 3)
- Firebase Cloud Functions (Phase 2 only if needed)
- WhatsApp order sharing for customers (Phase 2)

32. Rollout Plan

Phase 1 — MVP

Customer Features:

- Registration and login via WhatsApp OTP
- Product catalog with variants and Hinglish search
- Category browsing with horizontal tabs
- Cart with minimum order enforcement
- Order placement with pickup slot selection
- Order tracking and status updates
- Add items to pending order (before packing)
- Buy Again, Frequently Bought, Usual Orders
- Khata balance view, payment history, dispute flag
- Store info page

Owner Features:

- Store Open / Closed toggle
- Product and variant management
- Order management with Pack → Ready → Done workflow
- Customer label assignment (Trusted / Normal / Careful)
- Khata ledger management with credit limits
- Push notifications + WhatsApp fallback for new orders
- Owner note per order with customer binary response
- Store QR code and WhatsApp share link generation
- Abuse tracking via customer profile history

Phase 2

- Product images
- Language toggle (Hindi / Hinglish)
- WhatsApp order sharing for customers
- Low stock alerts

- Festive banners
- Basic analytics: top products, peak hours, cancellation trends
- Automatic customer behavior scoring (if owner wants it)
- Cloud Functions (only if specific need identified)

Phase 3

- UPI QR payment integration
- Delivery option (optional, within village)
- Reward coins and loyalty benefits
- Advanced loyalty campaigns

33. Risks & Mitigation

Risk	Mitigation
Fake / spam orders	Customer label system + max 1 active order rule
WhatsApp OTP unavailable	Fallback to Firebase email OTP or manual verification by owner
Owner misses order notification	WhatsApp fallback notification after 5 minutes
Customers forget to install PWA	QR code at store counter + WhatsApp onboarding link
FCM notifications disabled on device	WhatsApp fallback for owner; customer encouraged to allow notifications
Internet unavailable during order	Catalog browsable offline; error message guides retry when online
Firebase free tier limits hit	Daily limits: 50k reads / 20k writes — generous for 100 users. Monitor via Firebase console.
Owner overload	Simple UI — all actions in 2 taps or fewer
Customer forgets mobile number	Not applicable — everyone knows their own mobile number

34. Final Product Direction

This product is designed to function as: "A Smart Digital Dukaan Assistant for Village Retail"

Focused on:

- Repeat grocery orders with one-tap reordering
- Faster pickups with pre-packing workflow
- Customer trust through transparent khata and order tracking
- Digital khata replacing the physical notebook
- Simplified operations for a single owner on a smartphone
- Village-first usability with Hinglish, WhatsApp OTP, and QR onboarding
- Zero recurring cost — entirely on Firebase Spark free tier

35. Immediate Next Deliverables

Product Design

- Wireframes for customer app: catalog, cart, order status, khata
- Wireframes for owner dashboard: order cards, store toggle, customer profiles
- Hinglish UI copy library — all strings defined before build
- Mobile navigation patterns (bottom tab bar, back navigation)

Technical Design (Do Before Writing Code)

- Firestore data schema — especially product variants and order state machine
- Firestore Security Rules architecture
- Authentication flow with WhatsApp OTP provider
- Service Worker strategy for catalog offline caching
- FCM + WhatsApp notification architecture
- Store open/closed state management

MVP Build Priority Order

1. Authentication (WhatsApp OTP, session persistence)
2. Store open/closed toggle + store info page
3. Product catalog with variants
4. Cart with minimum order and pickup slots
5. Order placement, tracking, and state machine
6. Owner dashboard with order management
7. Push notifications + WhatsApp fallback
8. Khata system with credit limits and dispute flag
9. Customer label system (Trusted/Normal/Careful)
10. Repeat ordering features (Buy Again, Usual Orders)
11. Store discovery: QR code + WhatsApp share link